A One-Point Rigidification for Finite Lipschitz-Free Spaces

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Source And Open Problem

The source paper is Cuth–Doucha–Titkos, *Isometries of Lipschitz-free Banach spaces*, arXiv:2402.08266 [1]. In Section 8.2 it asks:

> 8.2. **Problems.** We find the following question the most pressing.
>
> **Question 1.** Does there exist a Lipschitz-free rigid metric space $\mathcal{M}$ such that $\bigcup E_{ext}(\mathcal{M})$ is not dense in $\mathcal{M}$?
>
> A very interesting answer to the previous question would be answering positively the next question. However, even a negative answer to the next question would be interesting.
>
> **Question 2.** Is $\mathbb{R}^d$, for $d \geqslant 2$, Lipschitz-free rigid? Either if it is equipped with the euclidean or with the Manhattan distance.
>
> **Question 3.** Does every metric space isometrically embed into a Lipschitz-free rigid space that contains only one additional point?
>
> REFERENCES

We address Question 3 in a finite graph-theoretic subcase. The source proves that every metric space embeds isometrically into a Lipschitz-free rigid metric space after adding three points. It asks whether one point always suffices.

Proof Intuition

The source's main rigidity criterion says that a weak Prague metric space is Lipschitz-free rigid once its preserved-extreme graph is 3-connected. If a finite metric space already has a 2-connected preserved-extreme graph, then adding one far-away point joined at the same distance to every old point turns that graph into its cone. Coning raises 2-connectivity to 3-connectivity. The large common distance prevents the new point from lying between old pairs, while every new radial pair becomes preserved extreme.

Result

For a finite metric space M, let $G_{\text{ext}}(M)$ be the undirected graph whose vertices are the endpoints of preserved extreme molecules and whose edges are the unordered pairs $\{x, y\}$ such that $m_{x,y}$ is a preserved extreme point of the unit ball of $\mathcal{F}(M)$.

Theorem 1 (One-point cone rigidification)**.** *Let M be a finite metric space such that $G_{\mathrm{ext}}(M)$ has vertex set M and is 2-connected. Fix $R > \mathrm{diam}(M)$, add a point $p \notin M$, and define a metric d_R on*

$$M_R = M \cup \{p\}$$

by keeping the original metric on M and setting

$$d_R(p, x) = R \qquad (x \in M).$$

Then M_R is Lipschitz-free rigid. In particular, M embeds isometrically into a Lipschitz-free rigid metric space with one additional point.

Proof. The formula defines a metric: for $x, y \in M$, one has

$$|d_R(p, x) - d_R(p, y)| = 0 \leq d(x, y)$$

and

$$d(x, y) \leq \mathrm{diam}(M) < R + R = d_R(x, p) + d_R(p, y).$$

Because M_R is finite, a molecule $m_{a,b}$ is preserved extreme exactly when no third point lies between a and b. This follows from the source's extreme-point criterion: in a finite metric space the positive gap in

$$d(a, z) + d(z, b) - d(a, b)$$

over all non-endpoint z is attained whenever $[a, b] = \{a, b\}$.

We now identify the preserved-extreme graph of M_R. If $x, y \in M$, then the new point p does not lie between them, since

$$d_R(x, p) + d_R(p, y) = 2R > \mathrm{diam}(M) \geq d(x, y).$$

Thus $\{x, y\}$ is preserved extreme in M_R exactly when it was preserved extreme in M. On the other hand, if $x \in M$ and $y \in M \setminus \{x\}$, then

$$d_R(p, x) = R < R + d(x, y) = d_R(p, y) + d(y, x),$$

and symmetrically $d_R(p, y) < d_R(p, x) + d(x, y)$. Hence no old point lies between p and x, so every pair $\{p, x\}$ is preserved extreme in M_R.

Consequently $G_{\mathrm{ext}}(M_R)$ is the cone over $G_{\mathrm{ext}}(M)$, with cone vertex p. A cone over a 2-connected graph is 3-connected: after deleting p and at most one old vertex, the old graph remains connected; if p is not deleted, then p is adjacent to every remaining old vertex. Therefore $G_{\mathrm{ext}}(M_R)$ is 3-connected.

The space M_R is finite, and its preserved-extreme graph has all points as vertices, so it is a weak Prague space. By Cuth–Doucha–Titkos Theorem 4.9 [1], every weak Prague metric space with 3-connected preserved-extreme graph is Lipschitz-free rigid. Hence M_R is Lipschitz-free rigid. □

Corollary 1. *Every finite 2-connected graph, equipped with its graph metric, embeds isometrically into a Lipschitz-free rigid metric space after adding one point.*

Proof. For a connected graph with its graph metric, the source Lemma 3.8 gives E_{ext} equal to the graph edge set. Apply Theorem 1. □

Verification And Limitations

The proof gives a one-point positive answer to Question 3 for finite metric spaces whose preserved-extreme graph is 2-connected. The construction is explicit: put the new point at any common distance $R > \operatorname{diam}(M)$.

This does not settle the source's full Question 3. The argument does not cover finite spaces whose preserved-extreme graph has cut vertices, and it does not treat infinite metric spaces. It also does not address Question 2 on $\mathbb{R}^d$.

Novelty check: the run indexes had no exact entry for arXiv:2402.08266. Web searches for the exact one-point rigidification wording and for "Lipschitz-free rigid" variants returned the source paper but no later solution. Local corpus search found no separate answer.

References

[1] M. Cuth, M. Doucha, and T. Titkos, *Isometries of Lipschitz-free Banach spaces*, arXiv:2402.08266, 2024. `https://arxiv.org/abs/2402.08266`